

FinTech Disruption in Financial Industry

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FinTech Disruption: Overview of Lecture

Part 1: Blockchain and Crypto

History of money and the Bitcoin genesis

Blockchain mechanics: hashing, consensus, smart contracts

Crypto market: \$3T peak, ETF approval, DeFi

CBDC: 134 countries, China e-CNY, EU Digital Euro

Part 2: Payments

Payment architecture: layered rails, RTGS, instant payments

FinTech disruptors: Stripe, Klarna, Wise, Alipay

Stablecoins and the Libra/Diem failure

Part 3: Credit and Lending

FICO limits and alternative data revolution

Marketplace lending: LendingClub, BNPL, SME credit

China P2P collapse and regulatory lessons

Part 4: Neobanks and InsurTech

Challenger banks: Chime, Revolut, Nubank

InsurTech value chain: telematics, AI claims, IoT

Central Theme

Technology lowers the cost of trust and intermediation — disrupting every layer of finance simultaneously.

Part 1: Blockchain and Cryptocurrency

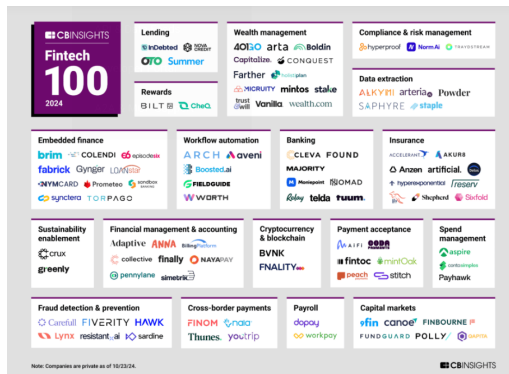
FinTech: Disruption Across All of Finance

Simultaneous disruption: FinTech is reshaping payments, lending, insurance, wealth management, and banking infrastructure at the same time

Entry barrier collapse: smartphones, cloud computing, and open APIs have reduced the cost to launch a financial product from \$100M to under \$1M

Competitive pressure: Big Tech (Alipay, Apple, Amazon) and focused startups attack different parts of the value chain simultaneously

Why now: the 2008 crisis destroyed trust in incumbents; a decade of near-zero rates funded experimentation; mobile penetration reached 6B+ users globally



The History of Money: From Shells to Satoshi

Aristotle's four functions: medium of exchange, unit of account, store of value, and standard of deferred payment — still the benchmark 2,400 years later

From barter to fiat: cowrie shells (Nigeria), Jiaozi promissory notes (Song Dynasty), private bank notes (US), and finally central bank fiat currency backed by legal mandate

Trust model: fiat money works because governments accept it for taxes and declare it legal tender — a social and political construct, not an intrinsic one

Digital money is not new: commercial bank deposits are already digital; the question is *who controls the ledger*



What Is the Role of Money?

Classical Foundations

Plato: money is a “symbol” for exchange; opposed gold/silver as money

Aristotle: money solves commensurability; enables future satisfaction of needs; must be durable, portable, divisible, and have intrinsic value

Modern additions: uniformity, acceptability, and supply stability

Network effects: money's value rises with adoption; fiat achieves this through mandatory tax acceptance and legal tender status

Forms Money Has Taken

Commodity money: silver dekadrachm (Greece), cowrie shells (Nigeria)

Representative money: Jiaozi note (Song Dynasty, 960 AD — world's first paper money)

Fiat paper: US dollar post-1971 (Nixon closed gold window)

Digital money: commercial bank deposits (\$90T+ globally)

Key Insight

Every shift in money's form was driven by a cost reduction: paper is cheaper than gold; digital ledgers are cheaper than paper.

Forms of Money Through History



Cowrie shells: Nigeria's currency for centuries; scarce, portable, durable

Jiaozi: Song Dynasty China promissory note, 960 AD — the world's first paper money issued by a government

Fiat paper: US Federal Reserve note; value backed by government decree, not gold

Private bank notes: US 19th century; thousands of competing bank-issued notes, varying in trustworthiness

Alipay mobile wallet: China 2004 — no physical form; purely ledger entries on Ant Group's servers

Fiat Currency: The Architecture of Modern Money

Three components: central bank notes (physical), central bank reserves (digital, held by commercial banks), and commercial bank deposits (held by individuals and firms)

Relies on ledgers: the entire money supply is a system of accounting entries; money “moves” by debiting one ledger and crediting another

Legal mandate: accepted for taxes and declared legal tender — creates powerful network effects; rejection is illegal

Bitcoin's question: can a purely mathematical system replace the trust that governments provide? Satoshi Nakamoto's 2008 paper said yes



Early Cryptographic Currencies: Why They Failed

Notable Attempts (1994–2000)

DigiCash (1994): David Chaum's blind signature system; technically elegant; went bankrupt 1998 — banks refused to adopt it

E-Gold (1996): gold-backed digital currency; reached \$2B in transactions; shut down by US DOJ for money laundering 2007

Bit Gold (1998): Nick Szabo's proof-of-work concept; never fully implemented; direct conceptual precursor to Bitcoin

B-Money (1998): Wei Dai's anonymous distributed electronic cash; cited in Satoshi's Bitcoin whitepaper

The Four Unsolved Problems

Double-spending: digital files can be copied; spending the same coin twice was trivially easy without a central authority

Centralization: all early systems relied on a trusted server — creating a single point of failure and a regulatory target

Merchant adoption: no network effects; merchants refused to accept without consumer demand; consumers refused without merchant acceptance

Consensus: agreeing on transaction order across untrusted parties (Byzantine Generals problem) had no efficient solution until 2008

Satoshi Nakamoto: The Bitcoin Whitepaper, October 31, 2008

The Context

Timing: the whitepaper was posted to a cryptography mailing list on Halloween 2008 — six weeks after Lehman Brothers collapsed and trust in banks hit a historic low

The problem solved: the *double-spending problem* — Nakamoto replaced a trusted intermediary with a mathematical proof (proof of work)

Byzantine Generals solution: nodes agree on the longest chain; an attacker would need 51% of computing power to rewrite history

Genesis block (Jan 3, 2009): embedded “The Times 03/Jan/2009 Chancellor on brink of second bailout for banks” — a political statement

“I’ve been working on a new electronic cash system that’s fully peer-to-peer, with no trusted third party.”

Core Innovation

Replace trust with math: instead of trusting a bank, trust cryptographic proof

Incentive design: miners earn Bitcoin for honest behavior; cheating is economically irrational

Identity: Satoshi Nakamoto’s true identity remains unknown; estimated to hold 1M BTC (\$60B+ at 2024 prices)

Blockchain Technology: Core Components

Cryptographic Primitives

Hash functions (SHA-256): produces a fixed 256-bit fingerprint of any input; changing a single character completely changes the hash — tampering is instantly detectable across every block in the chain

Merkle trees: summarize all transactions in a block into one root hash; allow lightweight “SPV” verification without downloading the full 500GB+ blockchain

Digital signatures (ECDSA): asymmetric key pairs prove consent without revealing the private key; only the private-key owner can authorize a spend — no username/password required

Consensus and Incentives

Proof of Work (Bitcoin): miners race to solve SHA-256 puzzles; consumes 127 TWh/year globally (comparable to Norway) — but makes cheating prohibitively expensive

Proof of Stake (Ethereum post-2022): validators lock up collateral; Ethereum’s “Merge” cut energy use by 99.9%, from 78 TWh to 0.01 TWh per year

Native token: block rewards make honest behavior economically rational; a 51% attack requires \$20B+ in hardware and still destroys the attacker’s own holdings

Key Design Principle

Every blockchain component solves the same problem: how to make cheating more expensive than cooperating — without any trusted authority.

How Blockchain Works: From Transaction to Finality

Step 1 — Broadcast: Alice signs a transaction with her private key and broadcasts it to the peer-to-peer network

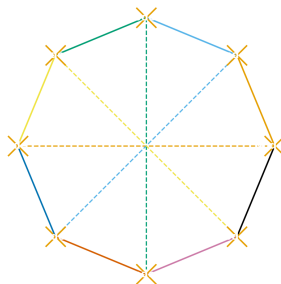
Step 2 — Validation: nodes verify the digital signature and that Alice has sufficient funds

Step 3 — Block formation: transactions are grouped into a candidate block with a timestamp and the hash of the previous block

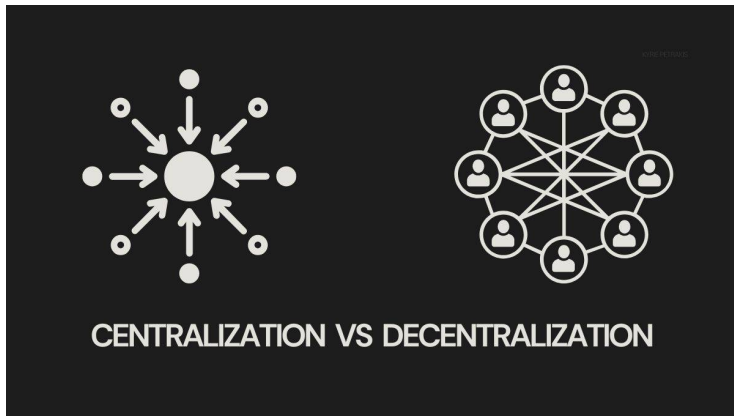
Step 4 — Consensus: miners (PoW) or validators (PoS) compete to add the next valid block; the first accepted by the majority wins

Step 5 — Finality: once 6 blocks are added on top (Bitcoin), the transaction is effectively irreversible

Peer-to-Peer (P2P) Blockchain Network



Centralized vs. Decentralized vs. Distributed Networks



Permissioned vs. Permissionless: Choosing the Right Architecture

Permissionless (Public)

Anyone can read, write, and validate: Bitcoin, Ethereum
— no gatekeeping

Strength: censorship-resistant; no single point of failure;
auditable by anyone globally

Weakness: slow (Bitcoin 7 tx/sec vs. Visa 24,000);
pseudonymous but traceable on-chain; governance by rough
consensus (hard forks are messy)

Best for: value transfer across untrusted global parties;
DeFi; sovereign assets

Permissioned (Private)

Approved members only: Hyperledger Fabric, Corda, JPM
Coin

Strength: fast, cheap, updatable, privacy-preserving

Best for: inter-bank settlement, supply chain, trade finance

The Blockchain Trilemma

Vitalik Buterin identified that any blockchain system
can optimize at most two of three properties
simultaneously

Security: resistance to attack; requires decentralization

Scalability: high throughput; requires fewer validators

Decentralization: no trusted center; requires more
validators

Key Question

Does the use case actually require multi-party consensus, or can a trusted intermediary do the job more cheaply? Most enterprise “blockchain” use cases are better served by a shared database.

Smart Contracts: Promise and Reality

What They Are

Nick Szabo (1996): “a set of promises, specified in digital form, including protocols within which the parties perform on these promises”

Ethereum’s contribution: Turing-complete scripts that execute automatically when conditions are met; no human needed to enforce

Applications: DeFi lending protocols (\$100B+ peak TVL 2021), NFT ownership, DAO governance, automated insurance payouts

Scale: Ethereum processes 1M+ smart contract calls per day (2024)

Critical Limitations

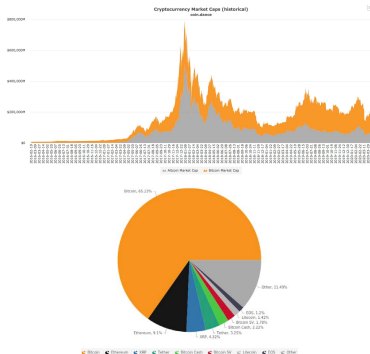
Not “smart”: code executes rules literally — no judgment, no intent, no learning; a bug executes as faithfully as correct logic (DAO hack 2016: \$60M stolen)

Not “contracts”: most lack offer, acceptance, and intent to be legally bound; courts cannot simply enforce a failed smart contract

Oracle problem: cannot see the real world directly; must rely on external data feeds (oracles) which are themselves trust points

Immutability trap: once deployed, code cannot be easily changed; upgrading requires complex proxy patterns

Crypto Market: Boom, Crash, and Recovery



Peak Nov 2021: total crypto market cap hit \$3 trillion; Bitcoin touched \$69,000; speculative frenzy across DeFi, NFTs, and meme coins

Crash 2022: market fell to \$800B; Terra/Luna algorithmic stablecoin collapsed (\$40B wiped in days); FTX fraud exposed (Nov 2022: \$8B customer funds missing)

Recovery 2023–24: Bitcoin surged past \$70,000 on BlackRock IBIT ETF approval (Jan 2024); IBIT attracted \$20B in assets within 60 days — fastest ETF launch in history

Institutional arrival: over 1,000 hedge funds and pension funds now hold crypto; Bitcoin's correlation with equities has declined, supporting its “digital gold” narrative

Source: CoinMarketCap, Bloomberg

Crypto Token Sectors: What the Market Actually Buys

Store of value (50–55% of cap): Bitcoin (\$1.3T+ as of 2024); institutional reserve asset; ETF approval is a structural demand shift

Platform tokens (20%): Ethereum (\$400B+), Solana (\$80B+), BNB; fuel for smart contracts and DeFi ecosystems

Stablecoins (6%): Tether USDT (\$120B+), USDC (\$35B+); essential for DeFi and crypto trading without converting to fiat

Source: CoinMarketCap (figures approximate, 2024)

DeFi tokens (5%): Uniswap, Aave, Compound; governance and fee-sharing in decentralized exchanges and lending pools

Real-World Assets (RWA): tokenized US Treasuries, private credit; \$10B+ in 2024; fastest growing sector

Meme coins and NFTs: speculative; high volatility; Dogecoin, Shiba Inu; NFT market collapsed from \$25B (2022) to \$2B (2024)

Blockchain Use Cases and Limitations

Genuine Use Cases

Cross-border payments: Ripple/XRP routes \$15B+ annually; 50%+ cheaper than SWIFT for some corridors

Trade finance: HSBC Contour reduced letter of credit processing from 5–10 days to 24 hours

Tokenized assets: BlackRock BUIDL fund (tokenized US Treasuries, \$500M+); Franklin Templeton on Polygon

DeFi: Uniswap processes \$1T+ in annual trading volume with no central entity

Hyped Use Cases (Unproven)

Supply chain: IBM Food Trust; most pilots failed due to data input trust problems (“garbage in, garbage out”)

Healthcare records: fragmented pilots; interoperability remains the real bottleneck

Structural Challenges

Scalability: Bitcoin 7 tx/sec; Ethereum 30 tx/sec; Visa 24,000 tx/sec; Layer 2 solutions (Lightning, Arbitrum) are closing the gap

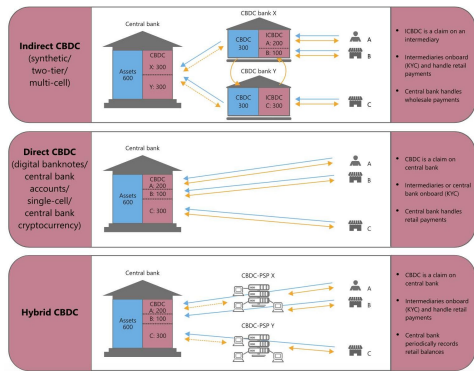
Privacy: all transactions on public chains are traceable by chain analytics firms (Chainalysis); not truly anonymous

Governance: protocol upgrades require consensus among miners, developers, and users; contentious splits produce hard forks (Bitcoin vs. Bitcoin Cash 2017)

Energy: Bitcoin consumes 127 TWh/year — comparable to Norway; Ethereum’s Merge (2022) cut its energy use by 99.9%

Central Bank Digital Currencies (CBDC)

CBDC Architectures: Who Holds the Liability?



Direct model: central bank holds every citizen's account; maximum government visibility and control; eliminates commercial banks from retail payments; privacy risk is severe

Indirect model: consumers hold bank deposits backed 1:1 by CBDC; commercial banks remain; closest to today's structure; least disruptive

Hybrid model: central bank holds CBDC liability directly, but private providers handle transactions; China's e-CNY operates this way — PBOC issues, commercial banks distribute

Design reveals values: every architectural choice is a political statement about privacy, financial inclusion, and state power

Source: BIS, "The Technology of Retail Central Bank Digital Currency"

CBDC: Global Progress and Real-World Results

Opportunities

Financial inclusion: unbanked populations hold digital money without a bank account; critical for sub-Saharan Africa and Southeast Asia

Monetary precision: programmable CBDC enables targeted helicopter money; interest-bearing CBDC as a new policy tool

Cross-border efficiency: BIS mBridge links 4 central banks (China, HK, Thailand, UAE) for instant FX settlement; bypasses SWIFT

Sovereign continuity: universally accepted risk-free means of payment; immune to private payment system failures

Real-World Progress (2024)

134 countries (98% of world GDP) are exploring CBDC (BIS tracker 2024)

China e-CNY: 260M+ wallets; \$250B+ in transactions; largest CBDC pilot globally; two-tier hybrid architecture

3 live launches: Bahamas Sand Dollar, Jamaica JAM-DEX, Nigeria eNaira — but Nigeria eNaira used by only 0.5% of population after 1 year

EU Digital Euro: ECB evaluation phase; potential 2028 launch; privacy vs. surveillance debate dominating hearings

CBDC Risks: Financial Stability and Surveillance

Financial Stability Risks

Bank run amplifier: CBDC provides a frictionless flight-to-safety option; in a crisis, citizens can instantly move deposits from commercial banks to government-guaranteed CBDC

Disintermediation: if citizens hold CBDC at the central bank, commercial bank deposit funding shrinks — reducing credit availability

Credit crowding-out: central banks may become direct lenders, displacing private credit allocation

Design Constraints

Most CBDC designs cap individual holdings (e.g., EU at €3,000) to limit bank disintermediation

Privacy and Governance Risks

Surveillance: government visibility into every transaction; China's e-CNY is traceable by PBOC but pseudonymous to merchants

Programmability: can governments expire money, geo-fence spending, or restrict purchases? US and EU face political backlash over this question

Cybersecurity: a CBDC system is a single point of failure for national payment infrastructure

Adoption: Nigeria's eNaira shows adoption does not follow automatically from launch

Key Tradeoff

Every CBDC design trades privacy for control. No country has found a solution that maximizes both.

Part 2: Payments

History of Payments: From Jefferson's Check to Instant Rails

Personal check (Thomas Jefferson, 1809): the first US personal check; paper-based deferred settlement; still 14B checks written annually in the US



Western Union telegram (1861): first electronic funds transfer; used Morse code and telegraph lines; dominated for a century



Telex (1950s–70s): teletype-over-telephone payment messaging; precursor to SWIFT for cross-border interbank communication



SWIFT (1973): replaced Telex; messaging standard for 11,000+ financial institutions; processes \$5T+ per day; but still T+1 or T+2 settlement

Check and telegram are in the public domain. Telex image by ajmexico on Flickr. CC BY.

Payment System Architecture: Layered Rails

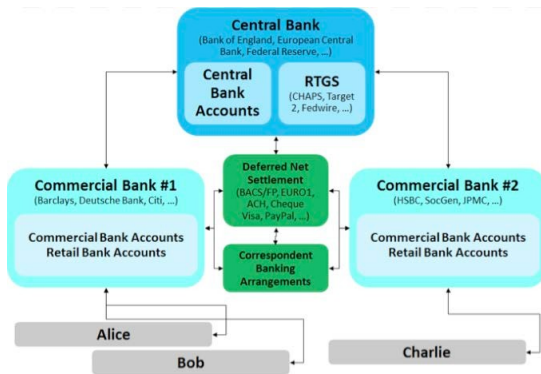


Image by Richard Gendal Brown. Used with permission.

Base layer: central bank reserves; the only risk-free money; settlement finality resides here; Fedwire, TARGET2, CHAPS

Middle layer: commercial bank accounts; money is a claim on the bank's reserves; banks net positions and settle at the central bank layer

Top layer: consumer apps and wallets — Venmo, PayPal, Apple Pay; these are claims on commercial bank deposits, not on the central bank

FinTech disruption target: most startups attack the top layer (UX, cost, speed) without touching the underlying settlement rails; FedNow and UPI are exceptions — they upgrade the middle layer

Modern Payment Ecosystem: Key Players

Consumer-Facing Wallets

Alipay / WeChat Pay: 1.3B and 900M users; embedded in super-apps; merchant fee 0.1–0.6% vs. 2–3% for Visa/Mastercard — total disintermediation of card networks

Apple Pay / Google Pay: NFC-based; layered on existing card rails; 500M+ global users; contactless penetration passed 30% of US transactions in 2023

M-Pesa (Kenya): 51M users; built financial infrastructure where banks never reached; 50%+ of Kenya's GDP flows through the platform

PayPal / Venmo: 430M+ accounts; Venmo 90M US users; under pressure from bank-owned Zelle (\$800B in 2023 volume)

Infrastructure and Instant Rails

Stripe (\$65B): developer-first APIs; powers Shopify, Amazon, Google; 135 currencies; processes hundreds of billions annually with 99.999% uptime

Adyen: enterprise-grade unified platform; processes for Netflix, Spotify, eBay; single API for online, in-store, and mobile

India UPI: 10B+ tx/month (2024); free for consumers; most successful instant payment system in history; 350M+ users in 8 years

FedNow (July 2023): 24/7 US instant payments; 800+ connected banks; closes US's 10-year gap vs. India and Brazil PIX

Key Insight

The winners in payments own the customer relationship. Alipay and WeChat Pay won by becoming the operating system for daily life — not just a payment button.

Global Payment Trends

Global eCom Payment Methods

	2019	2023*		2019	2023*
Digital/Mobile Wallet	41.8%	52.2%	Cash	30.2%	18.7%
Credit Card	24.2%	18.8%	Debit Card	24.3%	26.3%
Debit Card	10.6%	8.8%	Digital/Mobile Wallet	21.5%	29.6%
Bank Transfer	9.0%	9.3%	Credit Card	20.9%	22.3%
Charge & Deferred Debit Card	5.0%	3.0%	Charge Card	2.1%	2.1%
Cash on Delivery	4.5%	2.7%	Pre-Paid Card	1.0%	1.2%
Buy Now Pay Later	1.6%	2.8%			
PostPay	1.3%	0.9%			
Pre-Paid Card	1.0%	0.5%			
PrePay	0.6%	0.3%			
Other	0.6%	0.3%			

*Forecasted

Global POS Payment Methods

Credit card: dominant globally at 26% of POS value; contactless adoption accelerating

Debit card: 23% of POS; primary payment method in Europe and emerging markets

Digital/mobile wallet: 22% and rising fast; QR-code payments dominant in Asia

Cash: declining to 16%; still significant in developing markets; post-COVID collapse in advanced economies

Key trend: mobile wallets projected to overtake cards at POS by 2027

Source: Worldpay Global Payments Report

North America Payment Trends

North America eCom payment methods

	2019	2023*
Credit Card	34.4%	26.8%
Digital/Mobile Wallet	23.7%	36.6%
Debit Card	17.6%	15.2%
Charge & Deferred Debit Card	10.5%	9.1%
Bank Transfer	5.9%	6.3%
Cash on Delivery	3.2%	1.7%
PostPay	1.0%	0.2%
Other	1.0%	0.3%
Buy Now Pay Later	0.9%	3.0%
Pre-Paid Card	0.9%	0.3%
PrePay	0.9%	0.4%

North America POS payment methods

	2019	2023*
Credit Card	40.1%	41.6%
Debit Card	33.5%	31.5%
Cash	14.6%	10.8%
Digital/Mobile Wallet	6.0%	10.1%
Charge Card	3.9%	4.0%
Pre-Paid Card	1.9%	2.0%

*Forecasted

Card dominance: credit and debit cards account for 60%+ of US transaction value; contactless penetration surpassed 30% in 2023

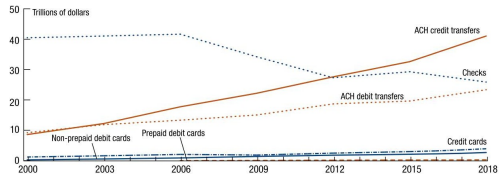
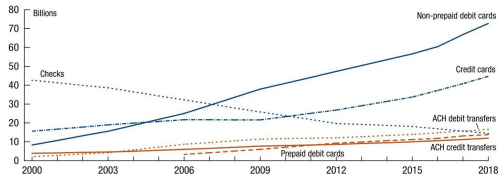
BNPL surge: Buy Now Pay Later (Affirm, Klarna, Afterpay) is the fastest-growing segment; capturing point-of-sale credit share from issuers

Cash collapse: post-COVID cash usage fell below 20% of all transactions; Sweden is approaching a near-cashless economy

Instant payment gap: the US was a decade behind; The Clearing House RTP (2017) and FedNow (July 2023) are now closing the gap with 800+ connected institutions

Source: Worldpay Global Payments Report

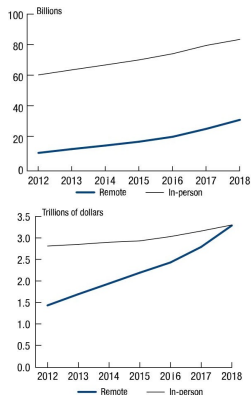
U.S. Payment Trends



Courtesy of the Federal Reserve and is in the public domain.

Source: 'The 2019 Federal Reserve Payments Study' (December 2019)

U.S. Card Payment Trends: The Interchange Economy



Scale: US credit card volume exceeds \$5T annually; debit cards \$4T+; together the largest single payment category

Interchange economics: merchants pay 1.5–3% per credit transaction; this funds rewards programs (airlines, cash back) and credit risk absorption

China's leapfrog: Alipay and WeChat Pay bypassed the card network entirely by linking directly to bank accounts; merchant fee 0.1–0.6% vs. 2–3% for cards

FinTech compression: Stripe, Square, and embedded finance players bundle software with payments — compressing fee revenue for traditional acquirers

Source: Federal Reserve Payments Study

Credit Card Economics: Where Does the Money Go?



In a typical **\$100 credit card purchase**:

Interchange (1.5–2.5%): paid to the issuing bank; funds rewards programs, fraud insurance, and credit risk; the largest slice and the biggest policy debate

Network fee (0.10–0.15%): Visa or Mastercard; covers authorization, clearing infrastructure, and brand value

Acquirer fee (0.20–0.30%): merchant's processor; terminal hardware, fraud detection, reporting

Net to merchant: approximately \$97.50; premium rewards cards can leave only \$96.50 — a significant cost for low-margin retailers

Why Alipay disrupted this: direct bank-to-bank transfer with 0.1–0.6% merchant fee; no card network, no issuer rewards — total disintermediation

Payment System Pain Points: Why Disruption Was Inevitable

Cost and Speed Problems

Cross-border payments: average cost is 6.3% of transaction value (World Bank 2023); correspondent banking chains add 2–5 day delays

Interchange: \$100B+ in annual merchant fees in the US alone; a direct tax on commerce that funds bank rewards

Delayed settlement: card networks settle T+1 to T+2; small merchants carry float risk on their own revenues

Chargebacks: merchants lose \$20B+ annually to fraudulent chargebacks; process favors cardholders

Access and Inclusion Problems

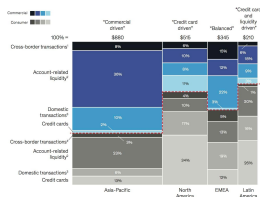
Unbanked: 1.4B adults globally lack a bank account (World Bank 2021); cannot access digital payments without a phone and bank link

SME exclusion: small businesses in developing markets cannot access card terminals or trade finance; M-Pesa and UPI have partially solved this

Data fragmentation: payment data is siloed across issuers, networks, and processors; open banking (PSD2 in EU, Section 1033 in US) is forcing sharing

Fraud: card-not-present (CNP) fraud exceeds \$15B annually in the US; rising with e-commerce growth

Global Payments Revenues: Scale and Shifting Geography



¹ Trade finance and cross-border payments services (B2B, B2C).

² Net interest income on current accounts and overdrafts.

³ Fee revenue on domestic payments transactions and account maintenance (excluding credit cards).

⁴ Remittance services and C2B cross-border payments services.

Source: McKinsey Global Payments Map

Total pool: global payments revenues exceeded \$2.2 trillion in 2023 (McKinsey); growing at 9% annually, driven by Asia-Pacific

Asia-Pacific dominance: China alone accounts for nearly 30% of global transaction volume; India's UPI is adding 10B transactions per month

Revenue shift: account-to-account (A2A) rails and instant payment schemes are eroding interchange-heavy card revenues; QR-code-based payment in Asia costs a fraction of card-based payment

B2B opportunity: the \$45B+ cross-border B2B payments market remains expensive (\$25–45 per transaction); prime target for Wise, Ripple, and SWIFT GPI

Source: McKinsey Global Payments Report 2023

Real-Time Gross Settlement and Instant Payment Rails

Large-Value RTGS Systems

Fedwire (1970): first RTGS; processes \$4T+ per day; the backbone of US dollar settlement

SWIFT (1973): financial messaging for 11,000+ institutions in 200 countries; \$5T/day in messages; not a settlement system — only a messaging standard

TARGET2 (EU), CHAPS (UK), CNAPS (China): national RTGS systems; typically operate limited hours

FedNow (July 2023): 24/7 instant retail payments in the US; 800+ institutions connected by end of 2024; maximum \$500,000 per transaction

Real-Time Retail Systems (24/7)

India UPI (2016): 10B+ transactions per month in 2024; free for consumers; the most successful instant payment system in history; 350M+ users

UK FPS (2011): first major real-time retail system; now processes 3B+ transactions annually

Brazil PIX (2020): 140M+ users in 3 years; free, instant, 24/7; eliminated checks and expensive TED/DOC transfers

US gap: US was 10 years behind India; FedNow is catching up but faces adoption inertia from banks

FinTech Payments: Two Waves of Disruption



Wave 1 — Big Tech (2004–2015)

Alipay / WeChat Pay: embedded in daily super-apps; near-zero merchant fees via direct bank account links; 2B+ combined users

Apple Pay / Google Pay: NFC device-based payments layered atop existing card rails; 500M+ users globally

Amazon Pay: leverages purchase history and 300M+ account trust

Wave 2 — Startup Infrastructure (2010–)

Stripe (\$65B): developer-first APIs; used by Amazon, Shopify, Google; 135 currencies; processes hundreds of billions annually

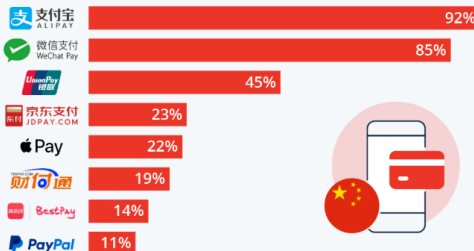
Klarna / Affirm: BNPL at checkout; shifting credit origination from banks to FinTech lenders

Wise / Revolut: low-cost cross-border transfers; Wise saves users \$1.5B in fees annually vs. banks

Global FinTech Payments Ecosystem

China's Most Popular Digital Payment Options

Share of Chinese online payment customers who used the following providers in the past 12 months



2,000 online payment customers from Mainland China (18-64 y/o)
surveyed Oct. 22-Sep. 23

Source: Statista Consumer Insights

Facebook Libra: Ambitious Design, Regulatory Collapse

Original Libra Vision (June 2019)

Global stablecoin: backed by a basket of G7 currencies; Swiss non-profit Libra Association with 28 founding members (Visa, Mastercard, PayPal)

Calibra wallet: custody and wallet services across WhatsApp and Messenger to 2.7B users

Goal: financial access for 1.7B unbanked people; near-zero-cost global transfers

Technical design: permissioned blockchain using Move language; Libra BFT consensus

Why It Failed (2019–2022)

Congressional hearing (July 2019): bipartisan hostility; “too big to trust”; moratorium demanded

Partner exodus: Visa, Mastercard, Stripe, PayPal, eBay all withdrew within weeks

Diem rebrand (2020): scaled back to single-currency stablecoins

Shutdown (January 2022): assets sold to Silvergate Bank; Move language lives on in Aptos and Sui

Key Lesson

Even a company with 3B users cannot override sovereign monetary authority. Regulatory legitimacy is a prerequisite for monetary infrastructure.

Stablecoins: The 2024 Landscape

Dominant Stablecoins

Tether USDT: largest at \$120B+; centralized, USD-backed; dominant in crypto trading and DeFi; reserve composition has been questioned by regulators

USD Coin (USDC): \$35B+; issued by Circle; preferred by institutions for transparency and compliance; monthly attestations of reserves

PayPal USD (PYUSD): launched August 2023; signals Big Tech re-entry into stablecoins after Libra failure; built on Ethereum

DAI: algorithmic, crypto-collateralized; decentralized; key DeFi building block; survived 2022 crash

High-Profile Failures

TerraUSD (UST): algorithmic stablecoin pegged algorithmically to Luna; collapsed May 2022; \$40B in market cap wiped in 72 hours; triggered broader crypto contagion

Facebook Libra/Diem: regulatory rejection ended project January 2022

Telegram GRAM: SEC blocked 2020; \$1.7B raised in ICO returned to investors

Wholesale Stablecoins

JPM Coin: JPMorgan permissioned stablecoin; \$10B+ in daily institutional transfers

Tokenized deposits: Citi, HSBC, BNY Mellon all piloting

Payment Companies Landscape

Card Networks and Legacy Processors

Visa (\$550B+ market cap): authorization, clearing, dispute resolution; 4B+ cards; 80M+ merchant locations

Mastercard (\$400B+): similar network; strong cross-border franchise; acquiring Vocalink (UK payments infrastructure)

American Express: closed-loop network; premium rewards; \$1.8T in annual billing volume (2023)

Incumbent PSPs: Fiserv (\$67B), FIS (\$35B), Global Payments (\$20B) — processing infrastructure for banks and merchants

FinTech Challengers

Stripe (\$65B): developer-first; powers Shopify, Amazon; processes hundreds of billions in annual volume

PayPal (\$65B): 430M+ active accounts; Venmo (90M users); Braintree (large merchants); facing competition from Apple Pay and banks' Zelle

Block / Square (\$40B): POS hardware + software + Cash App (56M users); vertical integration from payment to lending

Klarna (IPO 2024), Revolut (\$45B), Wise (\$9B): specialist disruptors in BNPL, neobanking, and cross-border respectively

Payment Industry Consolidation: The Megadeal Era

2019–2020 Incumbent Scale-Up Deals

FIS – Worldpay: \$34B (\$9B debt assumed); largest payment deal ever; created a \$100B+ processing giant; FIS later divested Worldpay at a loss in 2023 — a cautionary tale of overpaying at the peak

Fiserv – First Data: \$22B (\$17B debt); combined Clover POS hardware with processing infrastructure; more successful integration

Global Payments – TSYS: \$21.5B; consolidated mid-market processing

Worldline – Ingenico: \$8.6B; dominant European payment terminal company

Data and API Deals

Visa – Plaid (\$5.3B, blocked): DOJ antitrust challenge; Visa abandoned the deal 2021; Plaid independently valued at \$13B; would have given Visa control of open banking data aggregation

SoFi – Galileo (\$1.2B): banking-as-a-service infrastructure; powers Chime and other neobanks

Stripe – acqui-hires: Stripe has acquired 10+ smaller companies for team and technology

Key Pattern

Incumbents use M&A to buy scale and data access; regulators are blocking deals that would further concentrate payment infrastructure.

The Digital Currency Race: Private vs. Sovereign

		 中国人民银行 THE PEOPLE'S BANK OF CHINA
Name of currency	Libra	Digital Currency Electronic Payment (DCEP)
Issuer	Libra Association led by Facebook	People's Bank of China (PBoC)
Blockchain Type	Permissioned blockchain (accessible to association members only)	Centralized private network controlled by PBoC
Peg	Basket of currencies including USD, EUR, JPY, GBP and SGD	Renminbi (RMB)
Status	Release Date TBC	Release Date TBC

China e-CNY (DCEP): largest CBDC pilot globally; 260M+ wallets; used across 26 cities; \$250B+ in transactions by 2024; two-tier hybrid architecture keeps commercial banks in the distribution chain

Libra/Diem shutdown (Jan 2022): regulatory rejection by G7 governments demonstrated that private stablecoins at systemic scale are politically unacceptable without sovereign endorsement

US FedNow (July 2023): instant payments rail, not a CBDC; competes with Zelle and PayPal for real-time retail use; 800+ banks connected by end of 2024

EU Digital Euro: ECB evaluation phase through 2025; potential 2028 launch; privacy provisions under intense political debate

Source: BIS CBDC Tracker 2024

Part 3: Credit and Lending

Credit and Lending: Market Size and FinTech Entry

US Household Debt (\$17T+ in 2024)

Mortgages: \$12T+; FinTech mortgage origination (Rocket, UWM) is now 50%+ of US purchase originations by volume

Auto loans: \$1.6T; captive finance companies (GMAC, Ford Credit) remain dominant; FinTech underwriting used for non-prime

Student loans: \$1.77T; 90%+ federal; SoFi, Earnest compete in refinancing market

Personal loans: \$245B+; FinTech originates 40%+ by count — the most disrupted consumer credit segment

How FinTech Lenders Compete

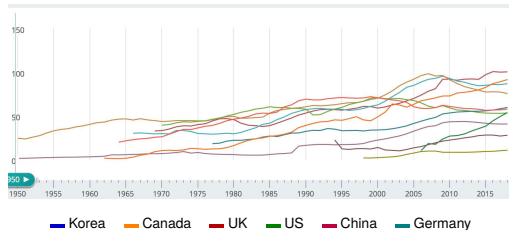
Data edge: alternative data (cash flow, app usage, behavioral) enables underwriting of borrowers FICO misses — 49M Americans are “credit invisible” or “thin file”

Speed: Upstart approves loans in 3 seconds; LendingClub in minutes; traditional banks in days to weeks

Funding models: balance sheet (OnDeck), securitization (SoFi), marketplace (LendingClub), bank partner (Affirm + USAA)

Channel: mobile-first, embedded at point of sale (BNPL), or API-first for embedded finance partners

Household Debt as Percent of GDP: Global Comparison



US and UK: among the highest at 75–85% of GDP; driven by large mortgage markets and consumer credit culture; mortgage debt alone is \$12T in the US

Korea and Canada: above 100% GDP; housing price surges drove record mortgage leverage; Bank of Korea raised rates aggressively in 2022–23 to cool household debt

China: rapid rise from near-zero (2000) to 60%+ GDP; primarily mortgage-driven; Evergrande default (\$300B) exposed systemic risk

Germany: low at ~55% GDP; strong renter culture; conservative borrowing preferences

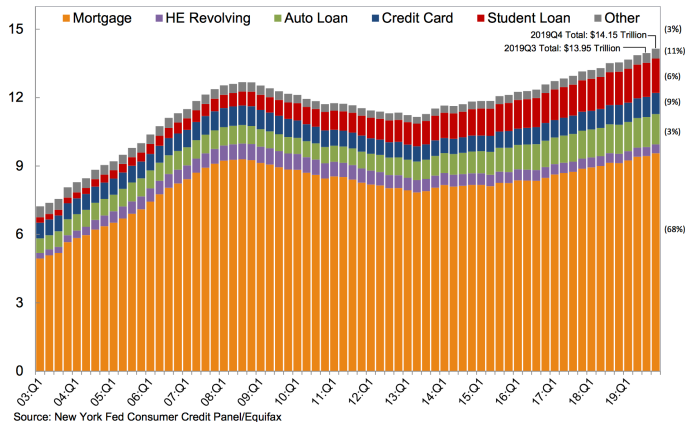
FinTech Context

High household debt drives demand for debt consolidation — the core use case for FinTech lenders: LendingClub (\$75B+ originated), SoFi, Upstart.

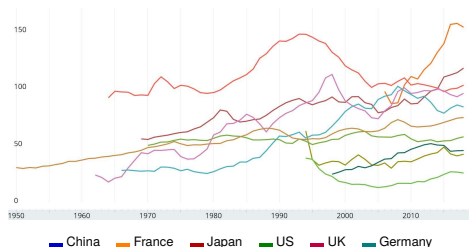
U.S. Household Debt Composition

Total Debt Balance and its Composition

Trillions of Dollars



Non-Financial Corporate Debt as % of GDP



China: highest at 160%+ GDP (2023); rapid rise driven by state-owned enterprise borrowing and property developer leverage; Evergrande (\$300B default) illustrates the systemic tail risk

France and Japan: elevated at 100–120% GDP; France's CAC-40 companies are highly leveraged; Japan's keiretsu groups use cross-shareholding and bank debt structurally

US: moderate at 80–90% GDP; 2020 COVID corporate bond wave added \$1T+ in IG issuance; investment-grade spreads near historic lows in 2024

Germany: lowest among G7 at ~60%; Mittelstand firms rely on retained earnings and Hausbank relationships rather than capital markets

FinTech Implication

High SME debt and limited bank access create demand for FinTech SME lenders: Funding Circle (\$15B+ originated), Kabbage, Square Capital.

FICO Scores: The Legacy System and Its Limits



FICO (Fair Isaac Corp., 1989): 35 years old; uses only 22 variables from credit bureau reports; designed for a world of branch banking and paper applications

Payment history (35%): single biggest factor; one missed payment can drop a score by 100+ points; disproportionately punishes young borrowers and immigrants

Credit invisible: 28M Americans have no FICO score at all; another 21M are “thin file” (score below 600); disproportionately people of color, immigrants, and young adults

FinTech response: Upstart uses 1,600+ variables including education, employment history, and cash flow; approves 27% more applicants at the *same default rate* as FICO models

Alternative Data in Credit: The New Frontier

Alternative Data Sources



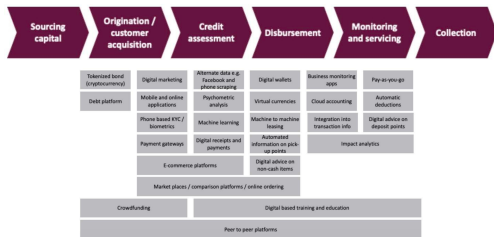
Cash flow underwriting: bank account transaction data is the strongest predictor of repayment ability; Plaid and Finicity aggregate this data for lenders in real time; JPMC Institute analysis finds cash-flow models approve 70% of gig workers rejected by FICO

Rental payment history: rent is often the largest monthly expense but excluded from FICO; Experian RentBureau and CreditLadder now report it; policy change could score 20M+ additional consumers

Open banking mandate: CFPB Section 1033 (finalized 2023) requires banks to share consumer financial data with any CFPB-authorized third party — leveling the data playing field

Regulatory risk: CFPB and FTC scrutinize algorithmic bias; ML models that use location or app data may proxy for protected classes under the Equal Credit Opportunity Act

Disruptive Technology Across the Credit Value Chain



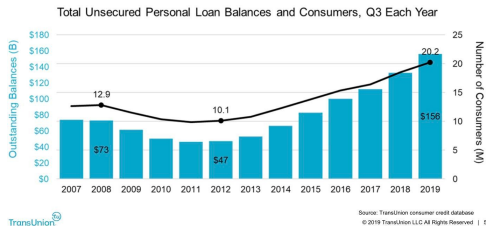
Origination: digital applications cut processing from days to 3 seconds (Upstart); mobile-first onboarding reduces drop-off rates by 50%+ vs. paper applications

Underwriting: ML models incorporating 1,000+ variables identify creditworthy thin-file borrowers that traditional banks reject; reduces both false positives (bad credit approved) and false negatives (good credit denied)

Funding: marketplace models (LendingClub) connect borrowers directly to institutional investors; securitization (SoFi) taps capital markets; eliminates bank balance sheet constraint

Servicing: AI-driven early delinquency prediction reduces charge-offs by 15–30% vs. rule-based systems; chatbot collections improve recovery rates at lower cost

U.S. Unsecured Personal Loans: FinTech Takes 40%



Record market size: US personal loan balances surpassed \$245B in 2023 (TransUnion) — fueled by debt consolidation demand and rising credit card APRs (averaging 21%+ in 2024)

FinTech market share: online lenders now originate over 40% of unsecured personal loans by count, up from less than 5% in 2013 — the fastest market share shift in consumer finance history

Speed advantage: online lenders offer approval in minutes and funding in 1 business day; traditional banks still take 3–7 days with branch visits

Stress test: FinTech delinquency rates rose sharply in 2022–23 as interest rates increased; ML models trained in a decade of low rates underestimated default risk in a rising-rate environment

Buy Now Pay Later: Rewriting Point-of-Sale Credit

How BNPL Works

Consumer perspective: split purchase into 4 equal installments (every 2 weeks); typically 0% interest if paid on time; no hard credit pull for pre-approval

Merchant perspective: pays 2–8% merchant fee; higher conversion rates (15–20% lift) justify the cost; BNPL providers assume the credit risk

Lender revenue: primarily merchant fees; consumer late fees and premium 0–36% APR installment plans for longer tenors

Market size: \$400B+ projected BNPL transaction volume globally by 2026 (Worldpay)

Key Players and Risks

Affirm: transparent APR; no late fees; Apple Pay Later partnership (2023); partnered with Walmart, Amazon

Klarna: “pay in 4”; Sweden-based; 150M+ users globally; IPO in 2024 at \$7B (vs. \$45B peak); squeezed by rising funding costs

Afterpay (Block): Australia-origin; acquired by Square/Block for \$29B (2022)

Hidden risks: BNPL users are 2x more likely to carry revolving credit card debt; CFPB issued guidance treating BNPL as equivalent to credit cards (2023)

LendingClub: The Marketplace Lending Pioneer



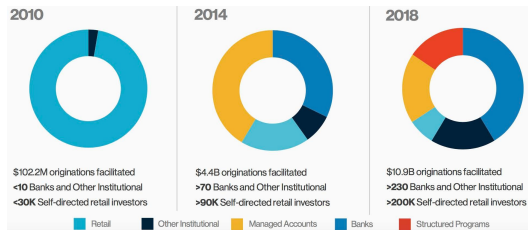
Originate-to-distribute: LendingClub originates loans online and sells them to institutional investors (hedge funds, banks, insurance companies); revenue is from origination fees (1–6%), not credit risk

Scale: \$75B+ in total originations since 2007; first FinTech lender to go public on NYSE (2014); NYSE: LC

Pivot to banking (2021): acquired Radius Bank — transforming from marketplace to bank holding company; shifted to retaining high-quality loans on balance sheet for interest income

Profitability: turned consistently profitable as a bank in 2022–24; demonstrates that FinTech lending requires either a stable funding base or a bank charter to survive a full credit cycle

LendingClub: Borrower and Investor Profiles



Typical borrower: prime/near-prime (FICO 680+); primary use case is debt consolidation (60%) — replacing 20%+ APR revolving credit card debt with a 10–15% fixed-rate personal loan

Investor evolution: historically dominated by hedge funds and banks buying tranches; retail investor participation has declined as institutional capital has professionalized the marketplace

Systemic risk: recession hits the entire loan portfolio simultaneously; no geographic diversification; correlated defaults unlike a branch-based bank with regional exposure spread

2024 position: profitable as a bank; credit tightening post-2022 improved loan quality; NIM (net interest margin) improved by retaining best loans on balance sheet

China Peer-to-Peer Lending: Explosive Growth and Collapse

The Boom (2012–2016)

2,680 P2P platforms at peak (2016); attracted by regulatory grey zone and underserved SME/consumer market

493B yuan (\$70B) outstanding at peak; investors earned 8–15%; platforms acted as guarantors without banking licenses

Structural flaw: platforms guaranteed principal — shadow bank balance sheets with no regulatory capital

Fraud wave: Ezubao (2016) stole 50B yuan from 900,000 investors; dozens of Ponzi schemes collapsed 2018–19

The Regulatory Crackdown

November 2019: strict P2P regulations requiring 1B yuan capital, institutional-only funding, or exit

Result: from 2,680 platforms (2016) to near-zero by 2021; \$140B in investor losses estimated

For regulators: guaranteed-return marketplace lending is shadow banking, not FinTech innovation

For incumbents: P2P collapse accelerated the pivot to Ant MYbank and WeBank with proper regulation

Key Lesson

Regulatory arbitrage is not a business model. Platforms that guarantee returns are banks, and must be regulated as banks.

FinTech SME Lending: Platform Data as Underwriting

Big Tech Platform Lenders

Alipay / Ant MYbank: AI approves SME loans in 3 minutes; \$290B+ outstanding (2023); default rates below traditional banks via real-time merchant data

Amazon: merchant cash advances auto-deducted from revenue; no application process

Square / Block: POS transaction history for underwriting; repayment fixed % of daily card receipts

Tencent WeBank: SME loans via WeChat ecosystem behavioral data

Specialist FinTech Lenders

Kabbage (American Express): working capital lines via bank account data; acquired by AmEx (2020, \$850M)

Funding Circle: UK-listed P2P marketplace; 80,000+ businesses funded; \$15B+ originated

Fundbox: invoice-based financing; API plugs into QuickBooks/Xero

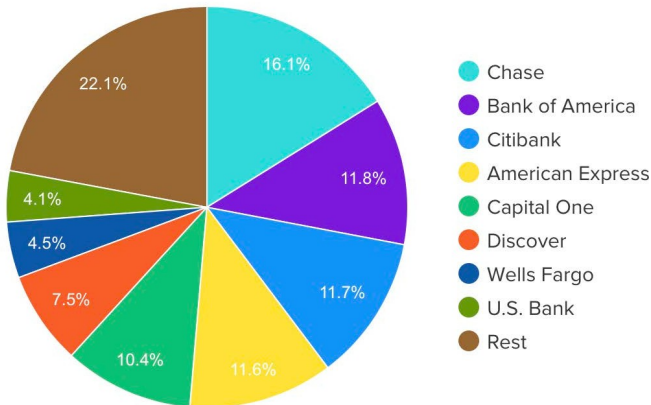
Brex: corporate cards for startups; underwriting based on funding and burn rate, not personal credit

Structural Advantage

Platform lenders see real-time revenue data, making credit decisions faster and more accurate than traditional annual-report-based underwriting.

FinTech Credit Landscape Overview

Market Share by Credit Card Issuer



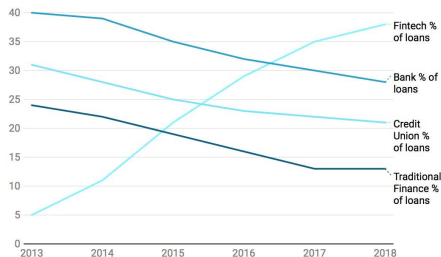
Largest Mortgage Lenders: FinTech Leads in Origination

THE WORLD'S 10 LARGEST MORTGAGE LENDERS BASED ON MARKET CAP			
Rank	Lender	Market cap	Head office location
1	JPMorgan Chase	\$563.54 billion	New York, USA
2	Bank of America	\$291.32 billion	North Carolina, USA
3	ICBC	\$246.51 billion	Beijing, China
4	Agricultural Bank of China	\$202.59 billion	Beijing, China
5	Wells Fargo	\$200.66 billion	California, USA
6	Bank of China	\$165.15 billion	Beijing, China
7	China Construction Bank	\$156.46 billion	Beijing, China
8	HSBC	\$152.25 billion	London, UK
9	Morgan Stanley	\$150.90 billion	New York, USA
10	HDFC Bank	\$150.11 billion	Mumbai, India

Top 10 largest mortgage lenders in the world by market cap

Personal Loan Industry Trends

Percentage of total unsecured personal loan balances originated by fintech vs. others



Source: TransUnion • [Get the data](#) • Created with Datawrapper

Market share shift: FinTech origination of personal loans grew from near zero in 2010 to 40%+ by 2022 — the fastest market share displacement in consumer finance history

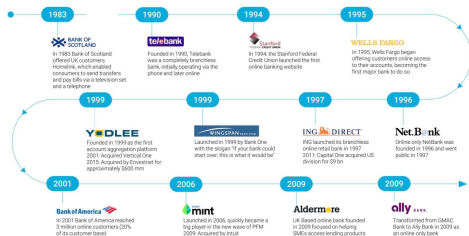
Cost advantage: digital lenders have 60–80% lower operating cost per loan vs. branch-based banks; no physical underwriting teams, no paper applications, no branch overhead

BNPL reshaping credit: Klarna, Affirm, and Afterpay are converting point-of-sale credit card purchases into installment loans — fundamentally changing how consumers conceptualize credit

2023 stress test: rising rates forced several FinTech lenders to tighten underwriting dramatically or seek bank partnerships; Upstart cut originations by 70%; Avant, Avant, and LendingClub all repriced risk upward

Part 4: Neobanks and Challenger Banks

From Branch to Pocket: History of Digital Banking



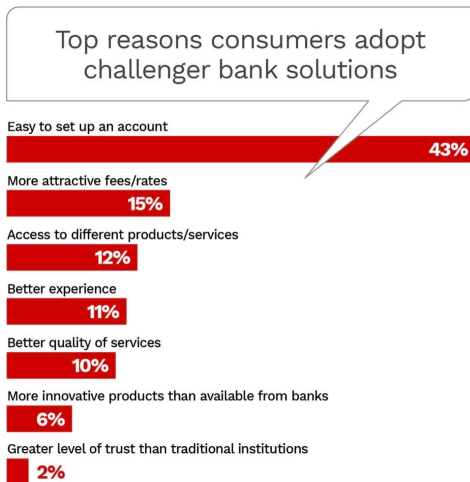
1994–2000: Wells Fargo and Bank of America launch first online banking portals; consumer adoption slow; security concerns; dial-up internet speeds limited usability

2007–2012: iPhone and Android transform banking into a 24/7 pocket service; mobile check deposit (USAA, 2009) eliminates a branch visit; mobile banking users surpass branch visitors by 2013

2014–2020: Monzo, Revolut, N26, Chime launch mobile-only with no legacy branches and no legacy IT; product is the app; acquisition via social referrals cuts CAC by 80%

2020+: COVID accelerates digital adoption by an estimated 5 years; branch visits fell 30%+ permanently; even elderly customers forced to adopt mobile banking; Bank of America Erica passes 1B interactions

Neobank Ecosystem Map



Challenges of Traditional Banking: Why Neobanks Exist

Structural Problems at Incumbents

Legacy technology: core banking systems (COBOL-based at many US banks) are decades old; adding a mobile app layer on top of a 1970s core creates friction, bugs, and slow feature delivery

Branch cost: a full-service branch costs \$2–4M annually to operate; US banks have 70,000+ branches; these costs are passed to customers as fees

Fees: average US bank account has 30+ potential fees; overdraft fee revenue alone is \$8B+ annually; targeted at lowest-income customers

UX gap: traditional banking apps rank far below neobanks in app store ratings; onboarding takes days vs. minutes

What Neobanks Offer

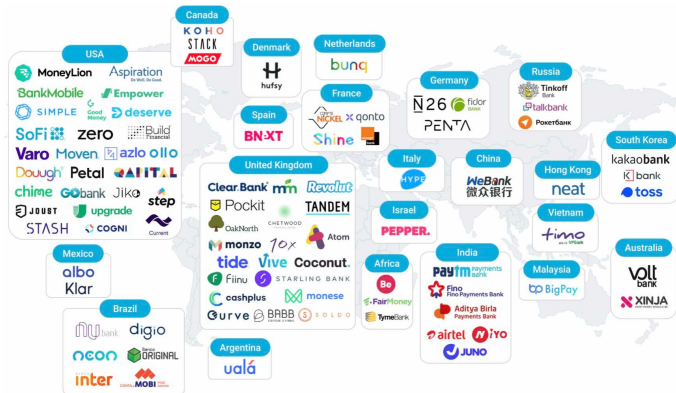
No fees: Chime, Revolut, and Monzo all launched with zero monthly fees; account opening in 5 minutes on a phone

Real-time notifications: instant push notification for every transaction; traditional banks notify via monthly statement

Early paycheck: Chime offers paycheck up to 2 days early via ACH timing arbitrage; key driver of growth for gig economy workers

International spending: Revolut and Wise offer interbank exchange rate with no markup on foreign currency transactions; banks typically charge 3–4% FX spread

Neo and Challenger Banks: Global Landscape



Courtesy of FT Partners. Used with permission.

Source: 'The Rise of Challenger Banks: Are the Apps Taking Over?', FT Partners (January 2020)

Neobank Product Offerings

Product	Revenue Stream	Example Companies (★ Indicates Starting Product Category)
 Checking / Savings Accounts & Debit / Prepaid Cards	• Interchange • Fee Income	 ★  ★  ★  ★  ★  ★   ★  ★  ★  ★  ★   ★  ★  ★  ★  ★
 Investment Products	• Fee Income	 ★  ★  ★  ★  ★  ★
 Installment Loans	• Interest Income (if on balance sheet) • Gain on Sale / Servicing Fees	 ★  ★  ★  ★  ★  ★  ★
 Student Loans	• Interest Income (if on balance sheet) • Gain on Sale / Servicing Fees	 ★  ★
 Credit Cards	• Interchange • Interest Income • Servicing Fees	 ★  ★  ★  ★  ★  ★
 Home Mortgages	• Interest Income • Gain on Sale • Servicing Fees	 ★  ★  ★
 HELOC / Home Equity Loans	• Interest Income • Gain on Sale • Servicing Fees	 (3) ★

(1) Coming in 2020
 (2) Currently only offered in Germany
 (3) Not offering complete challenger bank solutions, only home equity loans at the moment

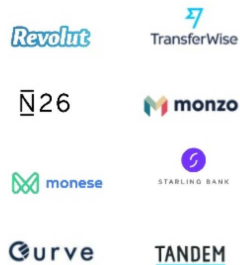
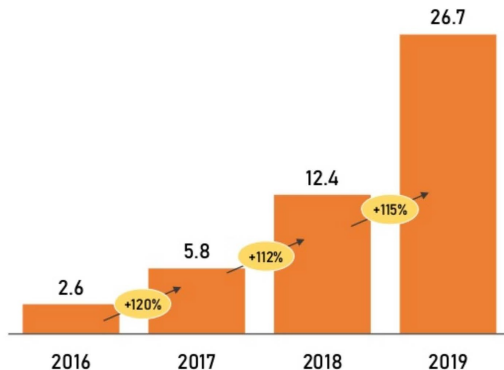
FT PARTNERS RESEARCH | 33

Courtesy of FT Partners. Used with permission.

Source: "The Rise of Challenger Banks: Are the Apps Taking Over?", FT Partners (January 2020)

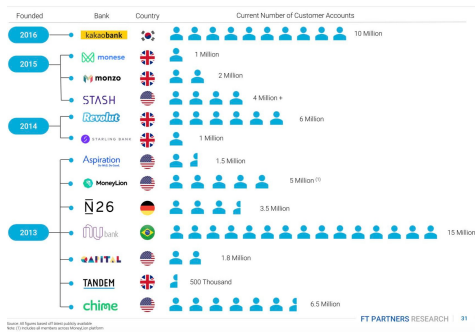
Challenger Bank Global Landscape

Number of customers – Selected top 8 bank challengers in Europe
In million



* including: Revolut, N26, TransfeWise, Monzo, Starling, Monese, Tandem, Curve
Source: Inteliace Research

Challenger Bank Customer Growth: Scale Without Profitability



CAC advantage: customer acquisition cost for neobanks is 5–10x lower than traditional banks (\$20–30 vs. \$300–500); driven by viral referral programs, social media, and word-of-mouth

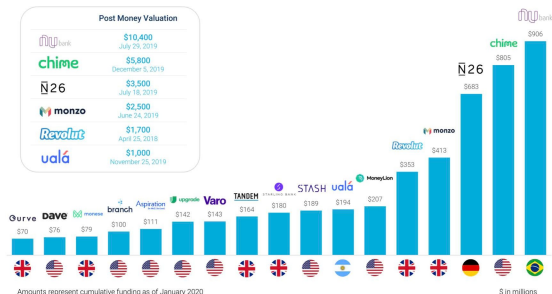
Chime (20M+ accounts): grew via direct deposit incentives and “get paid 2 days early” marketing; largest US neobank by accounts; primarily serves paycheck-to-paycheck customers

Revolut (45M+ customers): aggressive international expansion via a single EU banking license passported across 30 countries; fastest-growing financial app in history

Revenue gap: average annual revenue per user of \$15–50 for neobanks vs. \$200–400 for traditional banks; free accounts generate minimal income — monetization is the central challenge

Source: FT Partners, company filings

Neobank Funding and Valuations: Boom, Bust, and Reckoning



Source: FT Partners, CB Insights

Funding peak: neobanks raised over \$40B in VC globally between 2015 and 2022; concentrated in US, UK, and Brazil; investors bet on winner-take-all dynamics in digital banking

Valuation crash: Klarna raised at \$45B (2021) then \$6.7B (2022) — an 85% decline; Chime's \$25B peak valuation under pressure; IPO market shut for most neobanks through 2023

Path to profitability: Nubank (Brazil) turned profitable in Q4 2023 by cross-selling credit cards, personal loans, insurance, and investments to its 90M user base

The business model: acquire customers cheaply via free accounts, then monetize through premium plans, credit products, and investment services as the relationship deepens

Neobank Business Models Compared

Selected European bank challengers: Customers

PARTIALLY ESTIMATES

Company	Number of customers, <i>[no common definition]</i> in million, as of Dec. 2019*	Focus	Company valuation per customer post money, in '000 US\$, as of Dec 2019*
 Revolut	9.0	Accounts, fx, payments, P2P	0.82
 TransferWise	6.0	Payments, fx	0.58
 N26	4.5	Accounts, payments, business banking	0.78
 monzo	3.5	Accounts, payments, loans, business banking, other	0.74
 monese	1.4	Transfers, Accounts	1.03
 Starling	1.0	Accounts, payments, loans, business banking	n/a
 Curve	0.8	Payments, fx, card aggregation	0.40
 Tandem	0.5	Savings, credit cards	0.12

0.73
Average for top 7

Traditional Banks: The Mobile Counter-Offensive



JPMorgan dominance: Chase mobile app has 56M+ active users — larger than any neobank globally; annual tech budget of \$15B+; 3,000+ engineers working on digital products

Goldman Marcus: launched 2016; \$100B+ in deposits; Apple Card (12M+ cardholders); Goldman demonstrating that a bulge-bracket investment bank can compete in retail

Bank of America Erica: AI-powered virtual assistant with 1B+ customer interactions; demonstrates incumbent ability to deploy AI at scale with regulated data

Incumbent durability: trust, FDIC insurance, full-service lending, branch network for complex needs, and deep corporate banking relationships that neobanks cannot replicate quickly

Part 5: InsurTech

Insurance Value Chain: Where InsurTech Attacks



Distribution (highest disruption): online comparison platforms (PolicyGenius, Insurify, PolicyBazaar) cut broker commissions; customers bind policies in minutes; broker fees of 15–20% become indefensible

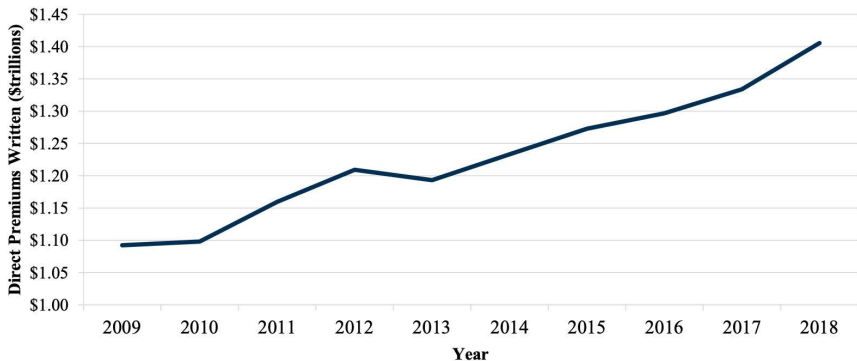
Underwriting: telematics (Root Insurance behavioral driving scores), IoT sensors (Vitality wearables linked to life premiums), and satellite imagery (property damage assessment) enable dynamic, individual-level pricing

Claims processing: AI computer vision (Tractable) assesses auto damage photos in seconds; drones inspect roofs post-hailstorm at scale; Lemonade pays some claims in 3 seconds via AI bot

Low disruption zones: reinsurance and capital markets remain dominated by incumbents (Munich Re, Swiss Re) due to massive capital requirements and centuries of actuarial data

Source: Federal Insurance Office, U.S. Department of Treasury

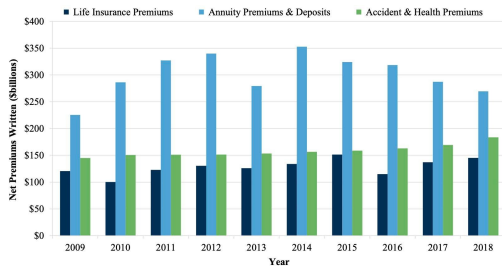
U.S. Insurance Sector Premiums



Source: S&P Global

Source: 'Annual Report on the Insurance Industry', Federal Insurance Office, U.S. Department of Treasury

Life and Health InsurTech: Innovation and Limits



Source: S&P Global

Market size: US life and health premiums represent 60%+ of total insurance premiums; health alone exceeds \$1.3T annually — the largest financial product most Americans buy

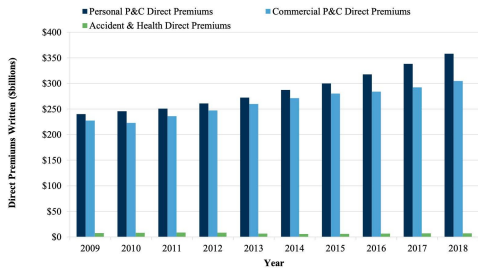
Life InsurTech wins: Ladder (term life in 5 minutes), Ethos (algorithmic underwriting eliminates medical exams for most applicants under 65); both use ML on health records and behavioral data

Health InsurTech struggle: Oscar Health and Bright Health both posted medical loss ratios above 100% in 2021–22; being an InsurTech does not eliminate actuarial risk

Biggest wins so far: distribution cost reduction (lower customer acquisition cost) and faster claims processing — not yet a fundamental transformation of underwriting accuracy

Source: Federal Insurance Office, U.S. Department of Treasury

Property and Casualty InsurTech: Telematics and AI



Source: S&P Global

Source: Federal Insurance Office, U.S. Department of Treasury

Primary battleground: auto and home insurance are commoditized products where price comparison and digital UX create the most immediate FinTech value; largest InsurTech funding concentration

Root Insurance: uses behavioral driving data (phone telematics) for pricing; went public 2020 (NYSE: ROOT); struggled with combined ratios above 120% in 2022 due to inflation and claims surge

Lemonade: renters and home insurance; AI claims bot pays renters claims in 3 seconds; \$2B+ market cap; still unprofitable — tension between growth velocity and unit economics

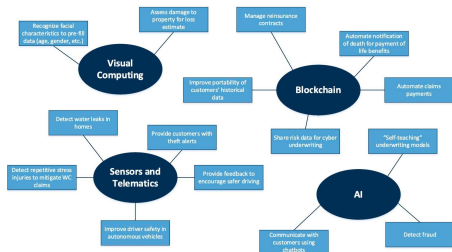
Climate risk limit: AI can improve pricing precision but cannot eliminate catastrophic correlated risk; major insurers are withdrawing from California and Florida markets

InsurTech Opportunities Map



Source: 'Quarterly InsurTech Briefing Q4 2019', Willis Towers Watson and CB Insights

InsurTech Emerging Technologies: IoT, AI, and Behavioral Data



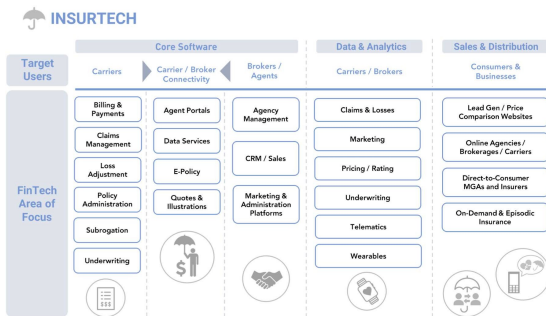
Telematics and IoT: connected car sensors score braking, acceleration, and phone distraction in real time; Progressive Snapshot has 20M+ users; Cambridge Mobile Telematics covers 200M+ miles analyzed daily

Wearables and health: Apple Watch linked to John Hancock Vitality plan; premium discounts tied to daily step counts and health metrics; \$150B wearables market generates continuous actuarial data

Drones: post-disaster roof inspection at scale; Allstate and Farmers deploy drone fleets after hailstorms; reduces adjuster deployment cost by 70%+ and speeds claim resolution from weeks to days

AI computer vision: Tractable processes auto damage photos in seconds; reduces total loss vs. repair decision time; processing 50%+ of auto claims for some insurers by 2024

InsurTech Funding: Boom and Bust Cycle



Source: FT Partners FinTech Almanac, CB Insights

Peak funding (2021): \$15.8B in global InsurTech VC investment; P&C received the largest share; health InsurTech is capital-intensive and regulation-heavy

Collapse (2022–23): funding dropped to \$3.7B in 2023 (CB Insights) — a 77% decline; IPO market closed; Root, Lemonade, Oscar all trade far below peak valuations

Structural problem: most InsurTechs remain unprofitable; capital consumed by marketing, licensing, and building actuarial reserves required by state insurance commissioners

Acqui-strategy: incumbents are increasingly acquiring rather than competing; Zurich acquired Cover Genius; Allianz invested in N26; Munich Re reinsures multiple InsurTech MGAs

InsurTech Competitive Map: Business Models Compared



Full-stack carriers (Lemonade, Root, Oscar): own the entire insurance risk; highest capital intensity; subject to full state-by-state insurance regulation; maximum upside if unit economics work; maximum downside in bad claims years

Managing General Agents (MGAs): Cover Genius, Pie Insurance — underwrite on behalf of reinsurers; lighter capital model; faster to launch; reinsurer bears the tail risk; margins are thinner

B2B technology platforms: Screenshot, Duck Creek, Applied Systems — sell software to incumbents; no insurance risk; steadier SaaS revenue; the “picks and shovels” of InsurTech

Data and analytics: Tractable, Cambridge Mobile Telematics, Shift Technology — sell AI models and data pipelines to carriers; capital-light; high-margin; growing rapidly as carriers digitize

Source: FT Partners FinTech Almanac

Key InsurTech Startups: P&C and Life

Property, Casualty, and Auto

Root Insurance (2015): telematics-based auto pricing; NYSE: ROOT; combined ratio above 100% in 2022; refocusing on profitable growth

Lemonade (2015): renters and home; AI bot “Maya” handles underwriting; 3-second claims; >\$800M raised; still posting losses on underwriting

Next Insurance (2016): small business P&C; digital-only; \$4B valuation (2021); targets sole proprietors and contractors underserved by traditional agents

Hippo (2015): smart home insurance; includes leak detectors; acquired spinnaker insurance carrier; SPAC-listed; valuation compressed sharply in 2022

Life and Health

Ethos (2016): life insurance; algorithmic underwriting; no medical exam for most; partners with carriers; \$2.7B valuation; 50% faster application than traditional

Ladder (2015): term life in 5 minutes; adjustable coverage without new underwriting; targets millennials who delay insurance purchases

Oscar Health (2012): individual and small business health; ML-driven care navigation; \$4B IPO 2021; improved to near-breakeven by 2023

ZhongAn (2013): China’s first online insurer; co-founded by Alibaba, Tencent, Ping An; listed Hong Kong 2017; 500M+ policies issued

Data-Driven InsurTech: The Real Revolution

Telematics and Driving Data

Cambridge Mobile Telematics (2010): 20M+ users; smartphone-based driving behavior scoring; powers usage-based insurance (UBI) programs for 60+ insurers globally

The Floow / TrueMotion: smartphone sensor data scoring; acquired by State Auto and Verisk respectively — showing incumbent acquisition appetite for telematics data

Habit Analytics (2018): behavioral patterns from smartphone and IoT data; identifies risk profiles without traditional credit scoring

AI Fraud and Claims

Shift Technology (2013): AI fraud detection for insurance claims; detects organized fraud rings and subtle claim inflation; \$220M raised; partners with 100+ carriers globally

Tractable (2014): computer vision for auto and property damage; processes photos in seconds; reduces total loss decisions from days to minutes; 3%+ market share of European auto claims

Snapsheet (2011): virtual claims management platform; enables fully remote claims processing; used by 130+ carriers

Key Insight

B2B data and technology providers are more consistently profitable than full-stack InsurTech carriers — they capture value without bearing actuarial risk.

Synthesis: Five Patterns Across FinTech Disruption

What Works

Attack the top layer: disrupting UX, cost, and distribution is achievable without owning the regulated core; Stripe, Wise, and Insurify prove this

Platform data advantage: lenders and insurers with proprietary behavioral data (Ant MYbank, Progressive Snapshot) outperform FICO/actuarial table-based competitors

Bank charter as endgame: LendingClub, SoFi, and Chime all moved toward bank charters or bank partnerships to solve the funding instability problem of pure FinTech models

What Does Not Work

Regulatory arbitrage: China P2P, TerraUSD, Libra all failed when regulators closed the gap between innovation and oversight

Growth without unit economics: Klarna (85% valuation drop), Root, Oscar, and Bright Health demonstrate that customer acquisition at any price is not a strategy in rising-rate environments

The Durable Insight

FinTech lowers the cost of intermediation. But it does not eliminate credit risk, insurance risk, or regulatory risk. The winners combine technological efficiency with financial discipline.